Mounika Darimireddy

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SUMMARY

- AI/ML Engineer and Data Scientist with seven years of experience designing, developing, and productionizing machine learning, cloud AI, and data-intensive Python solutions for complex enterprise requirements and high-volume enterprise data workloads.
- Experienced throughout the ML lifecycle, translating business objectives into analytical datasets, engineered features, modeling strategies, validation frameworks, deployable services, monitoring controls, and structured model-maintenance practices.
- Developed classification, regression, forecasting, clustering, anomaly-detection, and deep-learning solutions using rigorous experimentation, hyperparameter optimization, imbalance handling, explainability, and systematic error analysis.
- Engineered scalable data, feature, training, and inference pipelines using Python, SQL, PySpark, Apache Spark, Databricks, and Snowflake for distributed processing, cloud experimentation, and production scoring.
- Hands-on expertise in production GenAI and RAG applications using semantic retrieval, embeddings, vector databases, context engineering, LLM orchestration, prompt optimization, evaluation, guardrails, and source attribution.
- Implemented agentic AI workflows using LangChain and LangGraph, incorporating structured tool calling and MCP-based integrations for stateful routing, governed enterprise access, validation, fallback handling, and human escalation.
- Deployed scalable AI and ML services across AWS and Azure using FastAPI, SageMaker, Azure Machine Learning, Docker, Kubernetes, CI/CD, experiment tracking, model versioning, and production observability.
- Collaborated with product, data, application, cloud, and business teams to convert requirements into maintainable AI capabilities while contributing to technical design, production readiness, troubleshooting, and continuous improvement.

TECHNICAL SKILLS

Programming & APIs: Python, SQL, FastAPI, Flask, Django, REST APIs, Microservices, Object-Oriented Programming

Machine Learning & Data Science: Predictive Modeling, Classification, Regression, Forecasting, Clustering, Anomaly Detection, Deep Learning, Scikit-learn, XGBoost, TensorFlow, PyTorch, Feature Engineering, Model Validation, Hyperparameter Tuning, Statistical Analysis

Generative & Agentic AI: LLMs, RAG, Agentic AI, Amazon Bedrock, Anthropic Claude, LangChain, LangGraph, Pinecone, Vector Databases, Embeddings, Vector Search, Semantic Search, Metadata Filtering, Model Context Protocol (MCP), Tool Calling, Prompt Engineering, Context Engineering, Guardrails, RAG Evaluation

NLP & Document Intelligence: Natural Language Processing, Hugging Face Transformers, Text Classification, Named Entity Recognition, Text Summarization, Entity Extraction, Document Processing

Data Engineering, Big Data & Analytics: Pandas, NumPy, PySpark, Apache Spark, Databricks, Delta Lake, Snowflake, ETL, Data Ingestion, Data Transformation, Data Quality, Distributed Processing, Batch Processing, Feature Pipelines, Tableau, Power BI

Cloud Platforms: AWS (SageMaker, S3, Lambda, Glue, CloudWatch, IAM), Azure (Azure Machine Learning, Azure Databricks, Azure Data Factory, Blob Storage), GCP (Vertex AI, BigQuery, Cloud Storage)

MLOps & Deployment: MLflow, Model Registry, Experiment Tracking, Model Versioning, Model Serving, Model Monitoring, Drift Detection, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, Apache Airflow, Git

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Edward Jones - St. Louis, MO **Jan 2026 - Present**

- Delivered production-grade GenAI capabilities for an enterprise knowledge platform, combining RAG, semantic retrieval, and foundation models to provide grounded, source-backed answers from approved internal content.
- Partnered with product, data, application, and cloud teams to shape retrieval strategies, evaluation criteria, API contracts, and production controls, aligning GenAI capabilities with application requirements and accelerating delivery of reusable AI components.
- Created scalable ingestion and knowledge-indexing pipelines using Python and Amazon S3, processing structured and unstructured content through extraction, cleansing, chunking, metadata enrichment, embeddings, validation, and Pinecone indexing.
- Implemented Pinecone-based semantic retrieval with vector similarity search and metadata-aware filtering, returning query-relevant context while restricting results by document type, product area, knowledge source, and enterprise classification.
- Improved retrieval performance by tuning chunk size and overlap, similarity thresholds, top-K selection, metadata filters, and context limits, reducing irrelevant content and improving the evidence supplied to Claude.
- Designed query-processing and routing logic that normalized requests, applied retrieval criteria, and directed queries to retrieval, generation, validation, fallback, or escalation paths based on available context.
- Integrated Anthropic Claude through Amazon Bedrock and tuned prompts, retrieved context, inference settings, and response instructions to improve groundedness, consistency, completeness, and unsupported-query handling.
- Built reusable LangChain components for retrieval, context construction, prompt management, model invocation, output parsing, and response validation, simplifying integration and maintenance across RAG workflows.
- Orchestrated stateful agentic AI workflows with LangGraph and LangChain, coordinating retrieval decisions, tool selection, context preparation, response generation, validation, fallback handling, and human escalation.
- Extended agent workflows with MCP-based tools, allowing controlled access to approved enterprise APIs and knowledge services through structured tool calls, validated inputs, and predictable execution paths.
- Developed Python and FastAPI services for retrieval, agent orchestration, tool integration, and LLM inference, providing secure and reusable interfaces for downstream applications and enterprise services.

- Applied source attribution, confidence checks, Amazon Bedrock guardrails, output validation, and fallback controls to prevent unsupported responses when authoritative context was incomplete or unavailable.
- Evaluated RAG quality through retrieval relevance, context quality, groundedness, response completeness, latency, and failure-pattern analysis, using findings to target improvements across retrieval and generation.
- Automated incremental content processing and Pinecone reindexing for approved document changes, keeping searchable knowledge current without full index rebuilds or unnecessary service disruption.
- Deployed containerized AI services on AWS through Docker and CI/CD, applying IAM controls and CloudWatch monitoring to track latency, retrieval behavior, LLM interactions, workflow failures, and service health.

Machine Learning Engineer | Cognizant - Hyderabad, India **Mar 2019 - Dec 2023**

- Delivered end-to-end predictive analytics and machine learning solutions covering statistical analysis, model development, distributed data processing, cloud deployment, production integration, and post-deployment monitoring.
- Translated business requirements into structured machine learning problems by defining target outcomes, analytical datasets, feature strategies, modeling approaches, validation methods, performance criteria, and expected business impact.
- Engineered scalable, model-ready datasets using Python, SQL, Pandas, PySpark, Apache Spark, Databricks, and Snowflake, applying complex transformations, data-quality controls, feature generation, and optimized batch-processing patterns.
- Performed exploratory and statistical analysis to identify distributions, correlations, missing-data patterns, outliers, class imbalance, behavioral trends, leakage risks, and predictive relationships before model development.
- Created reusable preprocessing and feature-engineering pipelines covering missing-value treatment, categorical encoding, scaling, aggregation, feature construction, selection, dimensionality reduction, and consistent training and inference dataset generation.
- Developed classification, regression, forecasting, clustering, anomaly-detection, and neural-network models using Scikit-learn, XGBoost, TensorFlow, and PyTorch, selecting suitable techniques based on data characteristics and application requirements.
- Improved model stability and predictive quality through cross-validation, hyperparameter optimization, imbalance handling, threshold tuning, feature selection, explainability, baseline comparison, and systematic analysis of errors and edge cases.
- Developed NLP pipelines for text classification and named entity recognition using Python and Hugging Face Transformers, applying task-specific preprocessing, model validation, and systematic error analysis to unstructured data.
- Deployed batch and API-based inference across AWS and Azure using SageMaker, Azure Machine Learning, Python REST services, Docker, CI/CD, input validation, model packaging, and deployment-verification practices.
- Implemented reproducible MLOps workflows using MLflow experiment tracking, model versioning, artifact management, automated training, validation, batch scoring, candidate comparison, and controlled release practices.
- Monitored data quality, feature distributions, prediction behavior, model performance, drift, and service health, investigating degradation and supporting evidence-based maintenance, recalibration, and retraining decisions.
- Communicated model behavior, prediction drivers, evaluation findings, and business outcomes through explainability analysis, Tableau and Power BI dashboards, technical documentation, and stakeholder presentations.

Data Analyst | Amazon - Hyderabad, India **Oct 2018 - Feb 2019**

- Analyzed high-volume operational datasets using Python, Pandas, and advanced SQL, applying root-cause, trend, variance, segmentation, and exception analysis to isolate drivers of service-level, productivity, and quality performance.
- Engineered reusable analytical datasets by integrating, aggregating, and reconciling multi-source operational data while standardizing KPI definitions, business rules, and calculation logic for consistent performance measurement.
- Automated data extraction, transformation, validation, reconciliation, KPI computation, and recurring reporting through reusable Python and SQL workflows, improving processing efficiency, metric consistency, and reporting turnaround.
- Implemented data-quality and anomaly controls to identify missing, duplicate, inconsistent, and unusual records before analytical processing, increasing the reliability of operational metrics and downstream decision support.
- Communicated complex KPI trends, performance drivers, root causes, and data-quality findings through operational dashboards, analytical reports, and concise recommendations tailored to business and nontechnical stakeholders.

Python Developer | Oracle - Hyderabad, India **Jan 2018 - Sep 2018**

- Developed modular Python applications and automation utilities using object-oriented design, reusable components, and configuration-driven processing to streamline recurring data-intensive business and application workflows.
- Integrated Python applications with Oracle Database through parameterized SQL, PL/SQL routines, transaction handling, and reusable data-access components, enabling reliable data retrieval, validation, updates, and batch processing.
- Engineered Python, SQL, and PL/SQL batch and ETL workflows to ingest, cleanse, transform, validate, reconcile, and load multi-source relational data into standardized analytics-ready datasets.
- Optimized Python logic, data structures, database interactions, complex queries, stored procedures, and batch-processing patterns to improve application performance, processing reliability, and execution efficiency for recurring workloads.
- Embedded data-quality controls, exception handling, application logging, unit and functional testing, debugging, failure recovery, and technical documentation to support maintainable code and reliable production execution.

EDUCATION

Master of Science in Data Science | DePaul University, Chicago, IL **Jan 2024 - Nov 2025**

Bachelor of Technology in Computer Science | JNTUK, India **Jun 2014 - Apr 2018**